Durham University

**Wafer Bonding | CoMPhy Lab**
Department of Physics, Durham University
comphy-lab.org

CoMPhy Lab

# A table-top probe of wafer bonding: AFM and nanoscale gas drainage

Hybrid-bonded chips rely on two ultraflat silicon wafers snapping together while trapped gas escapes through a closing nanometre gap. Get the local drainage physics wrong, and the bond front leaves a void that kills the die. A wafer-scale bond event couples long-range elastic plate bending and an autonomous front-speed selection to a local drainage law in the closing nanometre gap; that local law is the closure every continuum bond-front model inherits (Navarro et al., 2013; Rieutord et al., 2005). This project isolates it. The single-asperity geometry is the minimal model: stripping the wafer-scale elasticity and autonomous front propagation out leaves the nanoscale squeezed-film law in the $\mathrm{Kn} \sim \mathcal{O}(1)$ regime, in its cleanest form. A heavily blunted atomic force microscope (AFM) tip, pressed against a polished substrate under controlled humidity, delivers a clean single-asperity realisation of that drainage law; axisymmetric direct numerical simulations of the same geometry close the loop.

**TL;DR**

You will run AFM force-distance experiments (with Dr Kislon Voïtchovsky, Durham) using a heavily blunted tip pressed against a flat substrate under a humidity sweep from dry nitrogen ($< 5\%$ RH) upward, and you will build direct numerical simulations (with Dr Vatsal Sanjay, CoMPhy Lab) of the squeezed film between tip and substrate. The two sides meet at a quantitative test of the nanoscale drainage constitutive law in the $\mathrm{Kn} \sim 1$ slip regime — the same sub-continuum piece that controls void formation and bond-wave distortion in industrial hybrid bonding at imec and TSMC (Nagano et al., 2022; Sato et al., 2025). This is the minimal model: the local drainage law, isolated from wafer-scale bending elasticity, that a Rieutord–Navarro bond-front solver needs as closure input.

## Description

Direct wafer bonding is the process by which two polished semiconductor wafers are brought into contact and seal together through van der Waals and capillary forces, with the ambient gas expelled radially as a bond wave propagates from an initial contact point (Gösele et al., 1995; Masteika et al., 2014; Plößl and Kräuter, 1999). The established front-speed scaling, derived by Rieutord and collaborators for flexible silicon wafers separated by a thin fluid layer, is $v \propto w^{5/4}$ with $w$ the wafer thickness (Rieutord et al., 2005); the most complete published model couples plate-bending theory to Reynolds lubrication and an adhesion-front boundary condition (Navarro et al., 2013). The industrial figures of merit are not overlay alone (overlay is a downstream alignment metric) but *void density, unbonded area*, and *bond-wave-induced distortion* (Nagano et al., 2022; Sato et al., 2025). Each of these is controlled by what happens inside the closing gap. As the gap approaches the molecular mean free path ($\lambda \approx 68\,\mathrm{nm}$ for air at STP), Knudsen slip, rarefaction, and sub-continuum gas dynamics all enter, and a naive continuum Reynolds closure drifts. Real closing gaps live at Knudsen numbers $\mathrm{Kn} = \lambda/h \sim \mathcal{O}(1)$: a regime that is neither continuum nor free-molecular, and where the appropriate closure is the Fukui–Kaneko generalised lubrication equation tabulated from DSMC-validated Poiseuille coefficients (Fukui and Kaneko, 1988, 1990).

The bond-wave physics is hard to probe on a production scanner — the gap is buried between two 300 mm wafers and closes in milliseconds. An AFM single-asperity geometry is the minimal model by construction: stripping out wafer-scale bending and autonomous front propagation isolates

the same nanoscale squeezed-film law that a continuum bond-front model takes as its closure. A standard AFM cantilever has a tip radius of $\mathcal{O}(10)$ nm; deliberately blunting it (Butt et al., 2005) to $\mathcal{O}(1\text{–}10)$ µm produces a hemispherical punch of known radius $R$, which can be driven against a polished substrate at controlled approach velocity $U$ under controlled humidity (Butt and Kappl, 2018). The force-distance curve carries three separable signatures: (i) pre-contact capillary jump-in at humidity-set separations (Israelachvili, 2011), (ii) squeezed-film viscous resistance that follows Stefan's law $F_{\mathrm{visc}} = 3\pi\eta R^2 U/(2h)$ (sphere-on-flat, thin-gap) at low $\mathrm{Kn}$ but drifts predictably as slip enters at $h \lesssim \lambda$, and (iii) a pull-off force set by JKR or DMT adhesion (Maugis, 1992), modulated by any condensed bridge formed during contact. The AFM sweep maps the local constitutive law $F(h; \mathrm{Kn}, \mathrm{Ca}, \mathrm{RH})$ — where $\mathrm{Ca} = \eta U/\gamma$ — not the front itself. The mapping to industrial bonding is indirect but concrete: once the local drainage law is quantified, it can be embedded as the closure in a bond-front solver of the Rieutord–Navarro class.

*Placeholder for two-panel schematic: (a) wafer bond-wave geometry (bending plates + lubricating gas layer + advancing adhesion front), (b) blunted-AFM single-asperity geometry isolating the nanoscale squeezed-film law that feeds (a).*

Figure 1: The AFM is a local constitutive probe, not a full bond-wave analogue. Panel (a) shows the wafer-scale problem: two bending plates, a lubricating gas layer, an autonomously propagating adhesion front. Panel (b) shows the single-asperity AFM geometry that isolates the thin-gap drainage law $F(h; \mathrm{Kn}, \mathrm{Ca}, \mathrm{RH})$ entering (a) as a closure.

This project closes that local loop. On the experimental side, you will run force-distance curves with a blunted AFM tip on silicon and oxide-terminated substrates, sweeping both approach speed (four decades, $10^{-9}$ to $10^{-5}\,\mathrm{m\,s^{-1}}$) and relative humidity (from $< 5\%$ dry-nitrogen purge upward in steps, so gas drainage and capillary condensation are separable in the data, not conflated (Butt et al., 2005)). On the theoretical side, you will build axisymmetric direct numerical simulations in Basilisk C (Popinet, 2015; Sanjay, 2022b) of the closing gap, with the Fukui–Kaneko slip-corrected lubrication closure for the gas phase to cover the continuum-to-rarefied transition (Fukui and Kaneko, 1988, 1990), and a disjoining-pressure potential $\Pi(h)$ with $\int \Pi\, dh = W_{\mathrm{adh}}$ as the molecular-adhesion closure. The experiment and the simulation meet at one non-trivial prediction: the speed and humidity dependence of the squeezed-film force at $\mathrm{Kn} \sim 1$. The tested closure then feeds back into bond-wave models of the Rieutord–Navarro class.

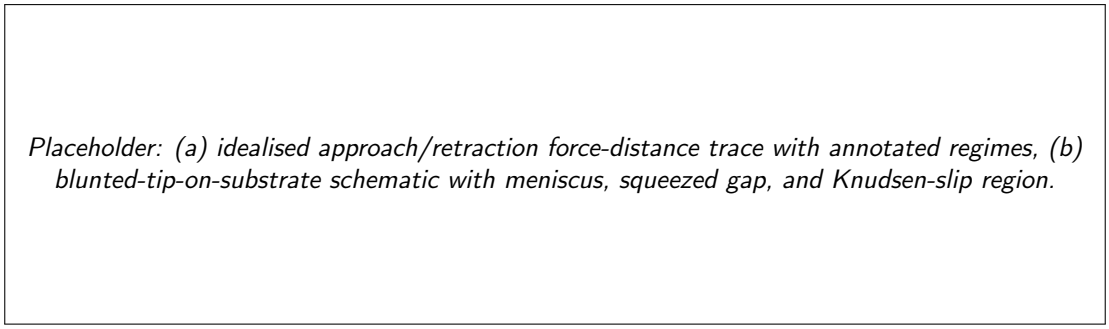



Figure 2: AFM force-distance signature of the three-force balance. Capillary jump-in (high RH, low $U$), viscous drainage (high $U$, continuum), and pull-off (adhesion-dominated retraction) are separable regimes of the same underlying model.

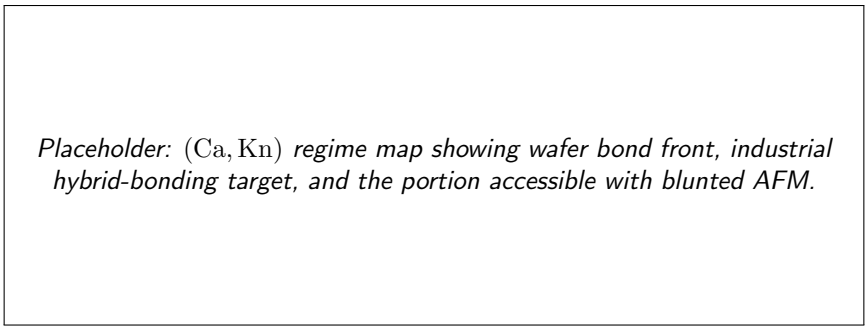



Figure 3: Dimensionless map. The blunted-AFM sweep covers the industrially relevant $\mathrm{Ca}$-$\mathrm{Kn}$ corner where continuum lubrication theory fails and sub-continuum gas physics sets the bond-front velocity. This is the target of the project.

## What you will do and what you will learn?

1. Learn force-spectroscopy AFM in Dr Voïtchovsky's lab: cantilever calibration, tip blunting protocols, humidity-controlled force-distance measurements, and data reduction from raw deflection to force.
2. Build an axisymmetric Basilisk C simulation of a blunted-tip approach to a flat substrate, with a slip-corrected Reynolds layer for the gas phase and a meniscus model for condensed water. Follow the CoMPhy-Lab two-phase and documentation conventions (Sanjay, 2022a,b).
3. Derive the local squeezed-film law in the three canonical limits — continuum Stefan ($\mathrm{Kn} \to 0$), Knudsen slip ($\mathrm{Kn} \sim 1$), and capillary-bridge dominated (high RH, low $U$). Test the Fukui–Kaneko generalised lubrication closure (Fukui and Kaneko, 1988, 1990) against the force-distance data, identify where it drifts, and bracket its quantitative validity range in $\mathrm{Kn}$. Place the measured $F(h; \mathrm{Kn}, \mathrm{Ca}, \mathrm{RH})$ alongside the established wafer-scale front-speed scaling $v \propto w^{5/4}$ (Rieutord et al., 2005), treating the AFM data as a direct measurement of the local drainage closure that Rieutord–Navarro-class bond-wave solvers require as input.
4. Position the results in the wafer-bonding landscape through the primary literature: the Rieutord 5/4 derivation (Rieutord et al., 2005), the Navarro coupled plate-bending / lubrication model (Navarro et al., 2013), the Gösele–Tong bond-speed analysis (Gösele et al., 1995), Masteika's review (Masteika et al., 2014), and the Nagano and Sato process-yield studies that actually quantify the industrial figures of merit (Nagano et al., 2022; Sato et al., 2025). Write

a short section of the final report on how the measured local drainage closure feeds a bond-front solver and which sub-100-nm physics it fixes for the wafer-scale problem.

5. Develop the CoMPhy-Lab full-stack open-source skillset: Basilisk, adaptive mesh refinement, Python post-processing, GitHub documentation, and writing for a combined soft-matter and semiconductor-physics audience.

## Industrial and scientific context

Hybrid bonding is the enabling technology for 3D-stacked logic, high-bandwidth memory, and back-side power delivery in the sub-2 nm node. TSMC's SoIC and Intel's Foveros both use it in volume. The dominant yield-limiting failure modes reported in the public process-engineering literature are not overlay error but *void formation*, *unbonded area*, and *bond-wave-induced in-plane distortion* (Nagano et al., 2022; Sato et al., 2025); overlay is the downstream alignment metric that these upstream failures corrupt. Each of these failures is seeded inside the closing gap, where sub-continuum gas drainage at Knudsen numbers around unity sets whether the front sweeps cleanly or pockets gas. Existing continuum-Reynolds closures in bond-front solvers of the Rieutord–Navarro class (Navarro et al., 2013; Rieutord et al., 2005) do not cover that regime; the Fukui–Kaneko generalised lubrication equation does, but it has not been directly benchmarked against a controlled single-asperity experiment in the relevant $\mathrm{Kn}$ range. The blunted-AFM setup is a clean-boundary-condition minimal model: it isolates the local closure and feeds it back to the wafer-scale solver. For a Durham MPhys or Master's project, this gives a crisp deliverable: a quantified drainage law at $\mathrm{Kn} \sim 1$, paired simulation and experiment, and an honest mapping onto a named industrial yield metric.

If you have any questions, feel free to contact us vatsal.sanjay@comphy-lab.org/ vatsal.sanjay@durham.ac.uk or drop by Ph255 (Rochester building) at the Department of Physics at Durham University.


| Collaborators | E-mail | Based at |
|---|---|---|
| Dr. Vatsal Sanjay | vatsal.sanjay@comphy-lab.org<br>vatsal.sanjay@durham.ac.uk | Ph255 (Rochester building) |
| Dr. Kislon Voïtchovsky | kislon.voitchovsky@durham.ac.uk | Dept. of Physics, Durham |
| Dr. Utkarsh Jain | utkarsh.jain@imec.be | imec, Leuven |


*Last updated: April 21, 2026*

## References


Butt, H.-J., Cappella, B., and Kappl, M. (2005). "Force Measurements with the Atomic Force Microscope: Technique, Interpretation and Applications". *Surf. Sci. Rep.* 59.1–6, pp. 1–152. DOI: 10.1016/j.surfrep.2005.08.003.

Butt, H.-J. and Kappl, M. (2018). *Surface and Interfacial Forces*. 2nd ed. Wiley-VCH.

Fukui, S. and Kaneko, R. (1988). "Analysis of Ultra-Thin Gas Film Lubrication Based on Linearized Boltzmann Equation: First Report—Derivation of a Generalized Lubrication Equation Including Thermal Creep Flow". *J. Tribol.* 110.2, pp. 253–261. DOI: 10.1115/1.3261594.

Fukui, S. and Kaneko, R. (1990). "A Database for Interpolation of Poiseuille Flow Rates for High Knudsen Number Lubrication Problems". *J. Tribol.* 112.1, pp. 78–83. DOI: 10.1115/1.2920234.

Gösele, U., Hopfe, S., Li, S., Mack, S., Martini, T., Reiche, M., Schmidt, E., Stenzel, H., and Tong, Q.-Y. (1995). "What Determines the Lateral Bonding Speed in Silicon Wafer Bonding?" *Appl. Phys. Lett.* 67.6, pp. 863–865. DOI: 10.1063/1.115530.

Israelachvili, J. N. (2011). *Intermolecular and Surface Forces*. 3rd ed. Academic Press.

Masteika, V., Kowal, J., Braithwaite, N. St. J., and Rogers, T. (2014). "A Review of Hydrophilic Silicon Wafer Bonding". *ECS J. Solid State Sci. Technol.* 3.4, Q42–Q54. DOI: 10.1149/2.007403jss.

Maugis, D. (1992). "Adhesion of Spheres: The JKR-DMT Transition Using a Dugdale Model". *J. Colloid Interface Sci.* 150.1, pp. 243–269. DOI: 10.1016/0021-9797(92)90285-T.

Nagano, F., Iacovo, S., Phommahaxay, A., Inoue, F., Chancerel, F., Naser, H., Beyer, G., Beyne, E., and De Gendt, S. (2022). "Void Formation Mechanism Related to Particles during Wafer-to-Wafer Direct Bonding". *ECS J. Solid State Sci. Technol.* 11.6, p. 063012. DOI: 10.1149/2162-8777/ac7662.

Navarro, E., Bréchet, Y., Moreau, R., Pardoen, T., Raskin, J.-P., Barthelemy, A., and Radu, I. (2013). "Direct Silicon Bonding Dynamics: A Coupled Fluid/Structure Analysis". *Appl. Phys. Lett.* 103.3, p. 034104. DOI: 10.1063/1.4813312.

Plößl, A. and Kräuter, G. (1999). "Wafer Direct Bonding: Tailoring Adhesion between Brittle Materials". *Mater. Sci. Eng. R* 25.1–2, pp. 1–88. DOI: 10.1016/S0927-796X(98)00017-5.

Popinet, S. (2015). "A Quadtree-Adaptive Multigrid Solver for the Serre-Green-Naghdi Equations". *J. Comput. Phys.* 302, pp. 336–358. DOI: 10.1016/j.jcp.2015.09.009.

Rieutord, F., Bataillou, B., and Moriceau, H. (2005). "Dynamics of a Bonding Front". *Phys. Rev. Lett.* 94.23, p. 236101. DOI: 10.1103/PhysRevLett.94.236101.

Sanjay, V. (2022a). *Code repository: Drop impact on viscous liquid films*. https://github.com/VatsalSy/Drop-impact-on-viscous-liquid-films (Last accessed: April 1, 2022).

Sanjay, V. (2022b). *Code repository: Impact forces of water drops falling on superhydrophobic surfaces*. https://github.com/VatsalSy/Impact-forces-of-water-drops-falling-on-superhydrophobic-surfaces.git (Last accessed: February 4, 2022).

Sato, R., Nagata, A., Kitagawa, H., Ogata, R., Myalitsin, A., and Inoue, F. (Mar. 20, 2025). "Factors Determining Bond Wave Speed in Wafer Bonding". *Jpn. J. Appl. Phys.* 64.3. Special Issue on Low Temperature Bonding for 3D Integration 2024 (LTB-3D2024), 03SP55. DOI: 10.35848/1347-4065/adbafc.